Beyond the Player: A Study of How Externalities in Baseball Affect Team Performance

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Abstract

Team performance in baseball and other sports not only relies upon the quality of the players, but also upon how those players are acquired. The most fundamental factors enabling player acquisition are the economic resources of the team. This study will principally investigate the relationship between the estimability of winning based on these economic factors. To define "winning," two approaches to measurement are taken: 1) by estimating the winning percentage of a team, and 2) by estimating whether the team makes the playoffs. The corresponding economic factors include payroll rank, market size, state income tax, prior season win percentage, stadium capacities, stadium rank, and Pythagorean win percentage. Rich in data, all thirty baseball teams are covered over a course of twenty-three years (1998-2022, 2020 being omitted) for a total of 7,920 analyzed data points.

Introduction

Just before the 2000's, the field of baseball saw a revolutionary change through the application of statistics, now widely known as sabermetrics. It was a new way of picking players and optimizing teams for the lowest cost, providing traditionally underdog teams a competitive edge.

The most famous example was in 2002, with the Oakland Athletics baseball team, for which a subsequent book and film titled *Moneyball* were produced. Ironically, the film's Wikipedia page has the best succinct description of what occurred:

> For the 2002 season the A's "[...] and their general manager Billy Beane attempt[ed] to assemble a competitive team with half (or even a third) as much money as his rivals. *Moneyball* follows Beane and his deputy Peter Brand scout the major leagues for undervalued talent by taking a sophisticated sabermetric approach to scouting and analyzing players [...]. Unable to afford more talented, expensive players, Beane and Brand focus on maximizing the team's on-base percentage (OBP) and compromise on skills like base stealing, defense, and batting average."
> (Wikipedia Contributors, 2024).

Though losing in the first round of division playoffs, the A's broke the American League's record for the most consecutive wins at 20, ending the season with a 0.64 win percentage, equivalent to a 103-59 record. This performance not only changed baseball but also introduced a completely new statistical approach to sports.

Once the Athletics noticed success in incorporating sabermetrics, teams began using money differently. The notion of a team receiving the greatest utility for their money is something inherently desired in sports. Now, they had a measurable way to achieve this goal. Since the Athletics implemented preliminary sabermetrics, teams have consistently improved at achieving the most success for their money spent. While once, the disparity between high and low-spending teams would be the single measure to predict success, analytics and sabermetrics have caused variables to become more ambiguous.

Due to the focus on sabermetrics, there is less attention given to the broader and often external factors that tip the tables, enabling a team to win. Payroll does still matter, but this factor can be complemented by where the team is located, the quality of the home stadium, average game day weather, and state income tax. The focus of this paper is to shed light on these disregarded variables, creating a more holistic picture of why some teams win while others lose.

Data and Variable Description

Among most professional sports, Major League Baseball has not only more games but also a longer season. Beyond being one of the oldest national sports in the United States, the magnitude of data also allows analyses to account for more recent structural changes in the game. Yet what determines whether a team wins or loses has changed over the decades. While the concept of using the relationship between payroll and winning to measure success has been studied many times, results generally conclude that a significant correlation does exist. One such analysis, conducted by Illinois Wesleyan University, ran a regression analysis in four distinct time frames ranging from 1977 to 2008. In each of these periods, they found a statistically significant positive relationship between payroll and regular-season winning percentage (Schwartz and Zarrow, 2009). In each of these periods, they also analyzed the correlation

between payroll and playoff wins. Yet this result was poor due to significant fluctuation, an R-squared of generally 0.00, and unexplained variance. Historically, it appears that a team's win percentage is more estimable than their chances of making the playoffs based on their payroll. But in terms of the World Series, it seems that teams good enough to make the playoffs tend to have about equal chances of winning the World Series, with the best team not significantly better than the worst at this cutoff.

Other studies find that prior-season winning percentage has a significant statistical correlation to team success in sports, indicating momentum plays a role in winning. A baseball-specific study, conducted by Jeff Sullivan for Fangraphs, who specializes in in-depth baseball statistics, found that there is a statistical correlation between first-half winning percentage and second-half winning percentage (Sullivan, 2018). Here, the R-squared is 0.27, which would typically be low but is relatively strong for a sports study. While no readily available study exists comparing one baseball season to the next, literature from other sports, as well as the significance of findings within this study, would support that the analysis should include a variable for prior-season win percentage within the model.

As hinted at, there are two established ways to measure success. Most literature measures team success through winning percentage in a given year. However, this is a short-sighted way to analyze what teams and organizations actually consider successful. Not only do fans care more about winning a World Series than the team's win percentage, but a team in a highly competitive division may not make it to the playoffs while one in a less competitive division does. To account for this discrepancy, two regression models will be utilized, each with its own distinct dependent variable:

I. Regular-Season Win Percentage — There have been numerous studies on whether money buys wins. In the interest of bridging the gap between the study's new model–with more explanatory variables–and theory, the decision was made to use regular-season win percentage as the first dependent variable. The regression results from this study can be compared with previous studies to verify that the study's results align with the literature. Note: Since OLS is a linear estimator and regular season winning percentage is bounded between 0 and 1, the study instead uses $ln(RegSeasonWin/(1 - RegSeasonWin)$–denoted "logwin"–to conform to OLS specifications.

II. Playoff Result — Teams often measure success through their ability to make the playoffs, which provides the opportunity to make a run for the World Series. So, when determining how to become a successful organization, teams make decisions in the interest of making the playoffs over winning percentage. Moreover, most public studies of success absent-mindedly analyze win percentage without considering what these wins actually represent. With this in mind, the alternative and more consequential dependent variable is the percentage of the time teams made the playoffs.

Note: Although it is expected that the results of these models are similar, the Playoff Result model is a better way to define the bounds for success. So long as the model for Regular-Season Win Percentage aligns with theory, it can be concluded that the explanatory variables, along with the included dataset used there, should work for the second model as well.

Next, it is important to consider what independent variables this analysis will use. It is reasonable to expect that payroll will likely have a positive correlation to regular-season winning percentage. Yet, this is an oversimplistic perspective on what factors go into success. Other studies have examined the effects of variables such as state income, market size, and stadium capacity on winning, with varying results. Instead of looking at these variables in isolation, this study wants to identify which variables constitute the strongest and most complete correlation with team success. In pursuit of this, consider the following independent variables:

I. Payroll Rank — Payroll is the amount a team spends in a given season on its players and related expenses, measured on the opening day of the season. Teams are given a rank, 30th for the highest spender and 1st for the lowest. This variable is expected to be significant and have a positive correlation due to this ordering.

Scatter Plot and Histogram of Payroll Rank

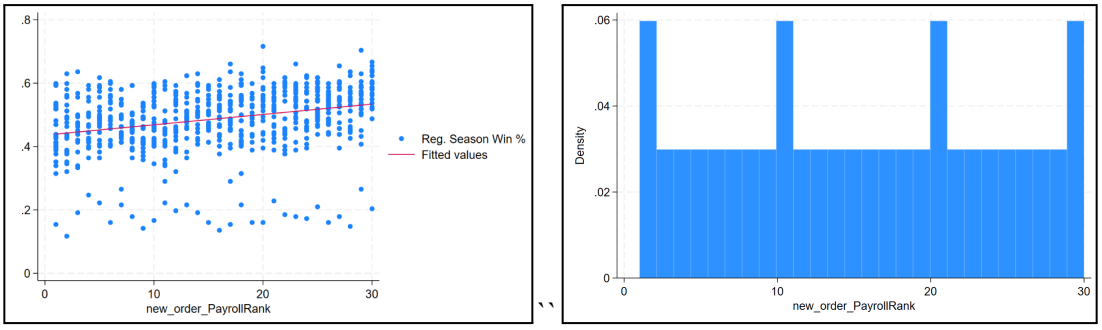


Figure 1: Scatter Plot and Histogram of Dispersion Payroll Rank Data

II. State Income Tax (base 2010, in decimal form) — 2010's highest tax bracket is utilized for each state because it is the median of this study's data observations (1998-2022), and no significant market disturbances were at play that would affect state income taxes. It is reasonable to assume players will choose states with lower state income taxes at comparable compensation. Therefore, it is expected that the coefficient for state income tax will be negative. Note that the province of Ontario was used as the "state" for the Toronto Blue Jays.

Scatter Plot and Histogram for State Income Tax

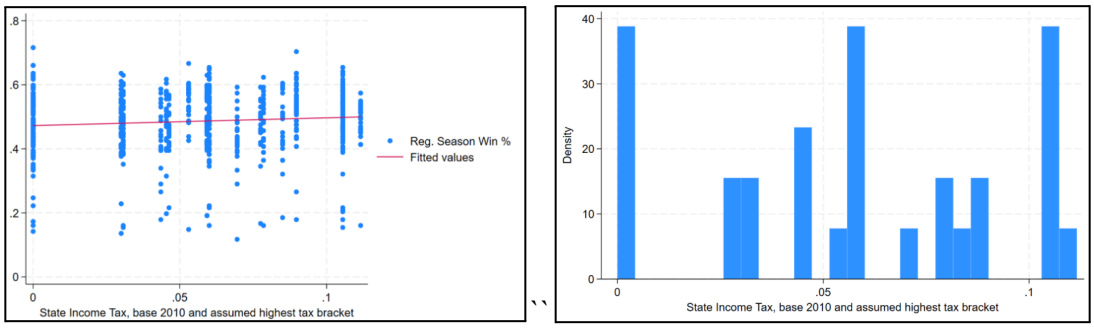


Figure 2: Scatter Plot and Histogram of Dispersion of State Income Tax Data

III. Market Size (in millions) — This variable utilizes the 2020 metropolitan statistical area for the location of the stadium. Theory generally states that market size should have a positive correlation to success. However, studies vary on the significance of the results, depending on the sample size and time series of the data. One study by Cork Gaines, who has authored studies for multiple sports entities, even found that the relationship between market size with both payroll and winning percentage is negligible (Gaines, 2011). Moreover, since the boundary for making the playoffs is lower than simply the most wins in a given season, it is reasonable to conclude that smaller market sizes may be more representative in making the playoffs than having the highest winning percentages, even if larger market sizes may dominate the highest win totals in a season. Therefore, the study does not hypothesize an expected coefficient for this variable, but includes it due to potential significance.

Scatter Plot and Histogram of Market Size

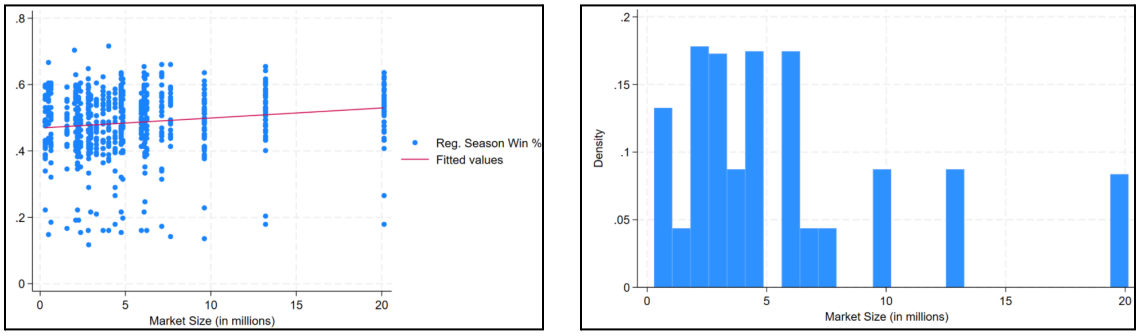


Figure 3: Scatter Plot and Histogram of Dispersion of Market Size Data

IV. Stadium Rank — This denotes a more qualitative factor that may still have some effect on the model. Stadiums are ranked from best being 30th and worst being 1st (for regression interpretive purposes) by consensus around ballpark aesthetic, experience, historicality, energy, etc. Players may do better when their ballpark has clout, contributing to increased wins and a greater chance at making the playoffs. Therefore, the expected coefficient for both models is positive.

Scatter Plot and Histogram of Stadium Rank

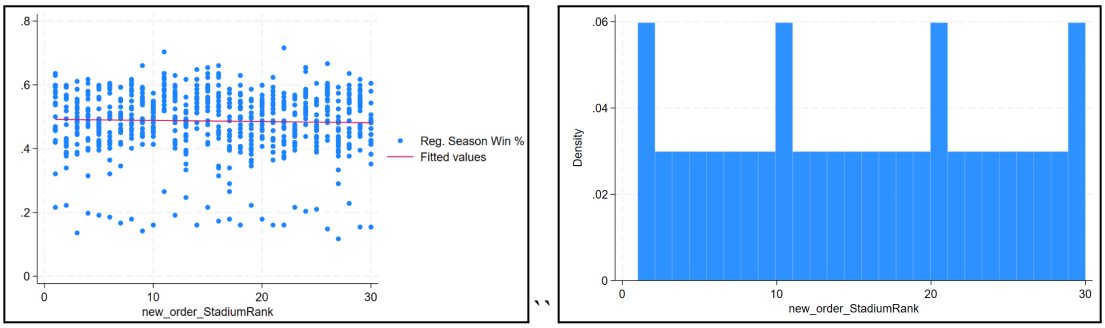


Figure 4: Scatter Plot and Histogram of Dispersion of Stadium Rank Data

V. Stadium Capacity — Teams that play at ballparks with larger stadium capacities may be indirectly or directly influenced. A positive coefficient is expected for this variable.

Scatter Plot and Histogram of Stadium Capacity

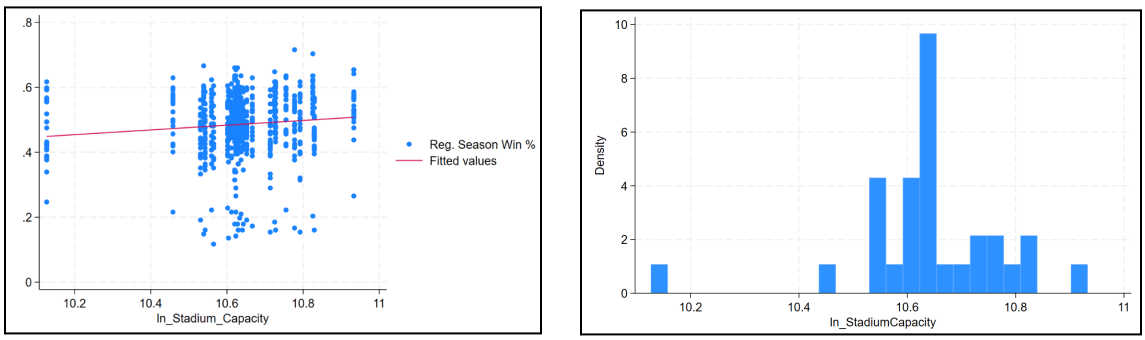


Figure 5: Scatter Plot and Histogram of Dispersion of the natural log of Stadium Capacity Data

VI. Prior Season Win Percentage — Calculated by dividing the previous season's wins by the total number of games (162). This variable is expected to have a positive coefficient because previous season performance carries weight over on how the team will do during the current season.

Scatter Plot and Histogram of Prior Season Win %

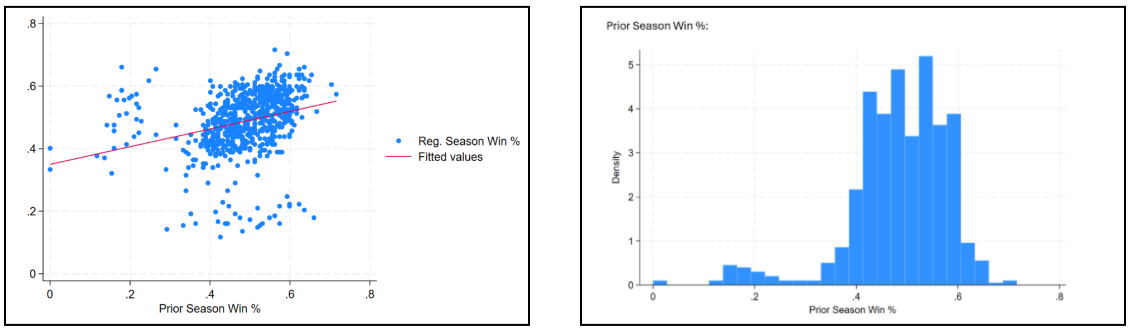


Figure 6: Scatter Plot and Histogram of Dispersion of Prior Season Win Data

Other variables were also considered, such as strength of schedule, runs scored, runs allowed, Pythagorean win percentage, lagged run differential, and weather. However, this info is not included in the primary study due to a few factors: a lack of data availability across the time range, data did not appear significant enough to include, or time constraints made it unrealistic to collect enough data in time while maintaining accuracy.

Interestingly, according to fundamentally every sports study ever conducted on the variables that affect a team's success, R-squared is always low—usually between 10% and 30%, depending on the study and variables used. Sports outcomes are inherently noisy and unpredictable, so models are unable to explain much of the variation. This might be attributed to:

- Randomness (luck) — Injuries and consistent bad umpiring decisions can heavily dictate the outcomes of a season for a given team.
- Missing variables — In sports, many variables are often hard or impossible to quantify, including team chemistry, coaching decisions, and team morale.
- Outcomes are compressed — Most teams generally win 70-100 games in a given season, with a winning percentage ranging from approximately .430-.620. The nuance between high and low values varies much less than many other types of data.
- Limited sample size — While MLB plays significantly more games than other sports, the datasets remain relatively limited.
- Competitive balance — Like in many other sports, the league is structured to reduce dominance and promote less-winning teams. One such example is the draft system, which tends to give teams that performed the worst in a given year the opportunity to draft the best prospects.

Together, these variables, while unable to be included in the model, will likely account for most of the variance in team success. Even if noise is unavoidable, the goal of this regression is to estimate the causal effect of included variables, not to capture all variation. Therefore, it is necessary to accept that every regression will only be partially complete.

The last aspect of the setup to explain is the data set. Much of sports theory is based on studies from relatively small sample sizes and relatively short time spans. In maximizing both of these central attributes, this study will maintain relevant data while increasing the reliability and accuracy of the estimates. To achieve this, the model uses fixed data regression from 1998 to 2022. The general starting point for most publicly available modern data is 1998, as well as the MLB's last expansion, which was at the beginning of the 1998 season, from 28 to 30 teams. 2020 was omitted due to irregularities caused by the COVID-19 pandemic and the shortened season.

All data used in this study was cumulated into one spreadsheet. The data is organized by column for each variable, with a total of 5040 relevant data points. The following spreadsheet and its data sources can be found in the references section of this paper.

Sample Spreadsheet Data

Current Team Name	Season	Payroll Rank	Playoffs Result	Adjusted Payroll	State Income Tax	Market Size	Stadium Capacity	Stadium Rank	Reg. Season Win %	Prior Season Win %	Pythagorean win %	Exotic (reg. -	Runs Scored	Runs Allowed
Baltimore Orioles	1998	1	0	111.887454	0.0695	0.284451	44970.00	4	0.488	0.605	0.519967064	-0.032	817	785
New York Yankees	1998	2	1	102.2383778	0.0897	20.14047	50287.00	20	0.704	0.593	0.683939243	0.020	965	656
Los Angeles Dodgers	1998	3	0	97.78998052	0.1055	13.200998	56000.00	7	0.512	0.534	0.493318784	0.019	669	678
Atlanta Braves	1998	4	1	93.54456	0.06	6.089815	41084.00	16	0.654	0.623	0.669004949	-0.015	826	581
Texas Rangers	1998	5	1	94.22900942	0	7.637387	40300.00	22	0.543	0.475	0.538045269	0.005	940	871
Cleveland Guardians	1998	6	1	92.70852885	0.05925	2.088251	34830.00	19	0.549	0.534	0.543502382	0.006	850	779
Boston Red Sox	1998	7	1	92.636829	0.053	0.494032	37755.00	5	0.568	0.481	0.590826883	-0.023	876	729
New York Mets	1998	8	0	91.33465541	0.0897	20.14047	41800.00	11	0.543	0.534	0.545059877	-0.002	706	645
San Diego Padres	1998	9	1	82.62385542	0.1055	3.298634	40209.00	6	0.605	0.469	0.581814844	0.023	749	635
Chicago Cubs	1998	10	1	77.5386	0.03	9.618502	41649.00	3	0.556	0.497	0.524015708	0.032	831	792
San Francisco Giants	1998	11	0	75.53741126	0.1055	4.749008	41265.00	2	0.549	0.556	0.566620852	-0.017	845	739
Los Angeles Angels	1998	12	0	75.341673	0.1055	13.200998	45603.00	23	0.525	0.543	0.502547754	0.022	787	783
Houston Astros	1998	13	1	75.209328	0	7.12224	40963.00	15	0.630	0.519	0.665237277	-0.036	874	620
Colorado Rockies	1998	14	0	74.29170694	0.0463	2.963821	50480.00	8	0.475	0.512	0.482753497	-0.007	826	855
St. Louis Cardinals	1998	15	0	68.64945968	0.06	2.820253	45494.00	17	0.512	0.451	0.517582501	-0.005	810	782
Seattle Mariners	1998	16	0	68.03799775	0	4.018762	47929.00	9	0.469	0.556	0.50233371	-0.033	859	855
Kansas City Royals	1998	17	0	55.44477	0.06	2.192035	37903.00	12	0.444	0.416	0.386796025	0.058	714	899
Chicago White Sox	1998	18	0	54.77526	0.03	9.618502	40615.00	28	0.494	0.497	0.460997014	0.033	861	931
Toronto Blue Jays	1998	19	0	53.1847845	0.1116	5.92804	41500.00	21	0.543	0.469	0.530275229	0.013	816	768
Milwaukee Brewers	1998	20	0	49.66503497	0.0775	1.574731	41900.00	24	0.457	0.484	0.431204295	0.026	707	812
Arizona Diamondbacks	1998	21	0	46.935765	0.0454	4.845832	48633.00	27	0.401	0	0.401450113	0.000	665	812
Philadelphia Phillies	1998	22	0	44.5652325	0.0307	6.245051	42901.00	13	0.463	0.42	0.437783804	0.025	713	808
Tampa Bay Rays	1998	23	0	42.61509	0	0.305275	25000.00	29	0.389	0.568	0.405313786	-0.016	620	751
Minnesota Twins	1998	24	0	38.1893175	0.0785	3.690261	38544.00	10	0.432	0.42	0.446034374	-0.014	734	818
Oakland Athletics	1998	25	0	38.1893175	0.1055	4.749008	46867.00	30	0.457	0.401	0.462925352	-0.006	804	866
Cincinnati Reds	1998	26	0	32.24131748	0.05925	2.256884	42271.00	26	0.475	0.469	0.493377774	-0.018	750	760
Detroit Tigers	1998	27	0	29.9527875	0.0435	4.392041	41083.00	14	0.401	0.488	0.411739477	-0.011	722	863
Miami Marlins	1998	28	0	23.574537	0	6.138333	37442.00	18	0.333	0	0.343062018	-0.010	667	923
Pittsburgh Pirates	1998	29	0	21.323115	0.0307	2.37093	38747.00	1	0.426	0.488	0.450414915	-0.024	650	718
Washington Nationals	1998	30	0	12.9231	0.085	0.638502	41339.00	25	0.401	0.481	0.403508379	-0.002	644	783
New York Yankees	1999	1	1	135.8864772	0.0897	20.14047	50287.00	20	0.605	0.704	0.602516735	0.002	900	731

Figure 7: Details the general format of data. Total of 721 rows and 15 columns for a total of 10815 points of data. Excluding the data needed to calculate regular season win % and pythagorean win %, as well as headers, there are 7920 relevant points of data.

Model and Empirical Strategy

Baseline Regression Model

$$Win\%_i = \beta_0 + \beta_1 PWP_i + \beta_2 PYR_i + \beta_3 SIT_i + \beta_4 MS_i + \beta_5 SR_i + \beta_6 SC_i + \varepsilon \quad (1)$$

$Win\% = team's\ regular\ season\ win\ percentage$

$PYR = payroll\ rank\ (1st\ being\ the\ team\ with\ the\ lowest\ payroll\ and\ 30th\ with\ the\ highest)$

$SIT = state\ income\ tax$

$MS = market\ size\ (in\ millions)$

$SC = stadium\ capacity\ (in\ thousands)$

$SR = stadium\ rank\ (1st\ being\ the\ best\ stadium\ and\ 30th\ being\ the\ worst)$

$PWP = previous\ win\ percentage$

Hypothesized Coefficient Signs

$$Win\%_i = \beta_0 + (+)\beta_1 PWP_i + (+)\beta_2 PYR_i + (-)\beta_3 SIT_i + (+)\beta_4 MS_i + (+)\beta_5 SR_i + \varepsilon \quad (2)$$

Estimated Functional Form

$$ln(Win\%_i) = \beta_0 + \beta_1 PWP_i + \beta_2 PYR_i + \beta_3 SIT_i + \beta_4 MS_i + \beta_5 SR_i + \beta_6 ln(SC_i) + \varepsilon \quad (3)$$

Note that the team's regular-season win percentage and stadium capacity are both in natural log form.

Ensuring Variables are Relevant

Testing For Heteroskedasticity

```
. estat hettest

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of RegSeasonWin

H0: Constant variance

    chi2(1) =   0.52
Prob > chi2 = 0.4704
```

Figure 8: Using Bruesh-Pagan Test to test for Heteroskedasticity.

When testing for heteroskedasticity using the above test, the desired outcome of the test is that there is little to no heteroskedasticity. The ideal result is that the Prob > chi2 is greater than 0.05, which this specific trial follows, suggesting little heteroskedasticity.

Testing For Multicollinearity

```
. vif

    Variable |       VIF       1/VIF
-------------+----------------------
new_or~lRank |      1.59    0.629282
MarketSize~s |      1.40    0.713451
ln~_Capacity |      1.39    0.718900
StateIncom~d |      1.18    0.847301
PriorSeaso~n |      1.18    0.847615
new_or~mRank |      1.08    0.927488
-------------+----------------------
    Mean VIF |      1.30
```

Figure 9: Using a VIF Test to test for Multicollinearity.

To test for multicollinearity, the best outcome is for the VIF to be below the value of 2.00. Fortunately, all the variables in the analyzed regression satisfy this test, suggesting little multicollinearity.

Testing For Correlation

```
. corr logwin StateIncomeTaxbase2010and new_order_StadiumRank new_order_PayrollRank MarketSize PriorSeasonWin ln_StadiumCapacity
(obs=720)

             |   logwin StateI~d ne~mRank ne~lRank Market~s PriorS~n ln_Sta..
-------------+---------------------------------------------------------------
      logwin |   1.0000
StateIncom~d |   0.0844   1.0000
new_or~mRank |  -0.0316   0.0413   1.0000
new_or~lRank |   0.2730   0.1827   0.2045   1.0000
MarketSize~s |   0.1579   0.2340  -0.0017   0.4557   1.0000
PriorSeaso~n |   0.2692   0.1144  -0.0105   0.3721   0.1490   1.0000
ln~mCapacity |   0.0971   0.3728   0.1404   0.3638   0.3930   0.0968   1.0000
```

Figure 10: Testing for Correlation between variables.

When testing for correlation between variables, the closer to 1 or -1, the more correlated two variables are to each other. It is ideal for correlation to be avoided between variables; therefore, the results of the test are desirable since the closest two different variables get to being correlated is 0.3930.

Tests for Heteroskedasticity and Multicollinearity were performed on each independent variable to ensure that every variable should be included. The tests for all the variables were satisfactory and led to the conclusion that none of the included variables violate “BLUE” and therefore can be used.

Results

Winning Percentage Model

Before running any analysis, the time spread of data should be accounted for. Data ranges from 1998 to 2022 (with 2020 omitted due to irregularities that year due to the COVID-19 pandemic). Therefore, the model should account for this in the regression analysis, as shown below.

Assigning Team ID’s

```
. xtset

Panel variable: team_id (strongly balanced)
 Time variable: Season, 1998 to 2022, but with gaps
         Delta: 1 unit
```

Figure 11: Creating team_id's allows implementation of panel data; id numbers range from 1-30.

From here, it is necessary to consider two methods: a Pooled Regression Model and a Fixed Effects Model. Determining the better option for the analysis requires a comparison of the regressions.

Fixed Data

```
. xtreg logwin StateIncomeTaxbase2010and new_order_StadiumRank new_order_PayrollRank MarketSize PriorSeasonWin ln_StadiumCapacity, fe
note: ln_StadiumCapacity omitted because of collinearity.

Fixed-effects (within) regression               Number of obs      =       720
Group variable: team_id                         Number of groups   =        30

R-squared:                                      Obs per group:
     Within  = 0.0498                                         min =        24
     Between = 0.3724                                         avg =      24.0
     Overall = 0.0844                                         max =        24

                                                F(5, 685)          =      7.18
corr(u_i, Xb) = -0.1194                         Prob > F           =    0.0000

------------------------------------------------------------------------------
       logwin | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
--------------+---------------------------------------------------------------
StateIncomeTa~0and |  -1.511678   1.084098    -1.39   0.164    -3.640232    .6168757
new_order_Stad~m |   .0003136   .0053848     0.06   0.954    -.0102592    .0108864
new_order_Payr~k |   .0086649   .0027757     3.12   0.002     .0032149    .0141149
MarketSizeinmi~s |   .0095114    .009892     0.96   0.337     -.009911    .0289337
PriorSeasonWin |   .6138145   .1692256     3.63   0.000     .2815514    .9460776
ln_StadiumCapacity |          0  (omitted)
             _cons |  -.4688619   .1513337    -3.10   0.002    -.7659955   -.1717283
--------------+---------------------------------------------------------------
           sigma_u |  .11428993
           sigma_e |  .39904719
               rho |  .07581036   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(29, 685) = 1.12                     Prob > F = 0.3103
```

Figure 12: Regression of RegSeasonWin vs. independent variables using a Fixed Data Regression.

This model exclusively focuses on changes within each team over time. These results show that payroll rank is a highly significant statistical predictor of winning ($p = 0.002$), indicating that when a team moves up in its payroll rank relative to its own history, it notices a statistically significant boost in winning. Prior-season win percentage is also a significant predictor ($p = 0.000$), demonstrating momentum may play a large role in year-to-year performance. Stadium rank is shown to be statistically insignificant ($p = 0.954$), likely since

stadium quality has relatively little change between years for a single team. Notably, stadium capacity is omitted entirely since it has no change for any teams in this sample. Regarding variance, the overall R-squared of this model is 0.0844, where a fixed-effects regression explains about 8.4% of the variation in the model. The most compelling part of this model specification is that the coefficients all align with the initial predictions, with state income tax being negative while all other coefficients remain positive. It should also be noted that the F-test (p=0.3103) may suggest that using this type of data is unnecessary.

Pooled Data

```
. regress logwin StateIncomeTaxbase2010and new_order_StadiumRank new_order_PayrollRank MarketSize PriorSeasonWin ln_StadiumCapacity
```

Source	SS	df	MS	Number of obs	=	720
				F(6, 713)	=	15.29
Model	14.6973613	6	2.44956022	Prob > F	=	0.0000
Residual	114.224565	713	.160202755	R-squared	=	0.1140
				Adj R-squared	=	0.1065
Total	128.921926	719	.179307268	Root MSE	=	.40025

logwin	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
StateIncomeTaxbase2010and	.2779365	.4538312	0.61	0.540	-.6130688	1.168942
new_order_StadiumRank	-.0034431	.0017895	-1.92	0.055	-.0069564	.0000701
new_order_PayrollRank	.0097539	.0021725	4.49	0.000	.0054886	.0140191
MarketSizeinmillions	.0030469	.0035203	0.87	0.387	-.0038644	.0099583
PriorSeasonWin	.8036077	.1647488	4.88	0.000	.4801569	1.127059
ln_StadiumCapacity	-.0200091	.1267614	-0.16	0.875	-.2688793	.2288612
_cons	-.3725218	1.333203	-0.28	0.780	-2.989996	2.244952

Figure 13: Regression of RegSeasonWin vs independent variables using a Pool Regression.

This model looks at the 720 observations as a single dataset, in effect comparing teams against each other. Payroll rank is still a highly significant predictor (p = 0.000). The same applies to prior season win percentage (p = 0.000). Stadium rank is now nearly significant at the 5% level (p = 0.055); however, it displays a negative coefficient. This would imply that teams with better stadium experiences tend to be associated with slightly fewer wins. It should be noted that this model can be more susceptible to omitted variable bias since it does not account for the unique identities of each team, possibly conflating stadium quality with other external factors. This might explain why other coefficients, such as stadium capacity (now included), misalign with the hypothesized signs. A comparison between the coefficients of each model is provided below. Even so, the R-squared of this model is 0.1140, indicating that about 11.4% of the variation is explained by the model. This is marginally higher than what the fixed effect model produced.

Fixed Data (m1) vs. Pool Regression (m2)

```
. estout m1 m2

------------------------------------------------------------
                              m1              m2
                               b               b
------------------------------------------------------------
StateIncom~d           -1.511678        .2779365
new_or~mRank            .0003136       -.0034431
new_or~lRank            .0086649        .0097539
MarketSize~s            .0095114        .0030469
PriorSeaso~n            .6138145        .8036077
ln~mCapacity                   0       -.0200091
_cons                  -.4688619       -.3725218
```

Figure 14: Comparing the signs of the coefficients from the results of a Fixed Data Regression versus a Pool Regression. Result: Use Fixed Data Regression since coefficient values match theory.

Final Comparison and Model Selection

While the Pooled model offers a slightly higher Adjusted R-squared and retains static variables such as stadium capacity, the fixed-effects model is the preferred choice for this research. Even if the F-test suggested it was not strictly required, the Fixed-Effects model is a more conservative approach that is less likely to incur omitted variable bias, while maintaining hypothesized coefficient signs that align with previous literature and logical reasoning. Moreover, the exemption of stadium capacity does not detract from the model's primary finding: that year-over-year performance is derived from payroll fluctuations and past success. Even when stadium capacity is included in the pooled data model, it is not statistically significant and therefore less relevant to the model than other explanatory variables.

"logwin" Dependent Variable Regression

```
. xtreg logwin StateIncomeTaxbase2010and MarketSizeinmillions PriorSeasonWin new_order_StadiumRank new_order_PayrollRank ln_Stadium_Capacity
> , fe
note: ln_Stadium_Capacity omitted because of collinearity.

Fixed-effects (within) regression               Number of obs      =       720
Group variable: team_id                         Number of groups   =        30

R-squared:                                      Obs per group:
     Within  = 0.0498                                         min =        24
     Between = 0.3724                                         avg =      24.0
     Overall = 0.0844                                         max =        24

                                                F(5, 685)          =      7.18
corr(u_i, Xb) = -0.1194                         Prob > F           =    0.0000

------------------------------------------------------------------------------
      logwin | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
StateIncome~0 |  -1.511678   1.084098    -1.39   0.164    -3.640232    .6168757
MarketSizei~s |   .0095114    .009892     0.96   0.337    -.009911    .0289337
PriorSeason~n |   .6138145   .1692256     3.63   0.000     .2815514    .9460776
new_order_S~k |   .0003136   .0053848     0.06   0.954    -.0102592    .0108864
new_order_P~k |   .0086649   .0027757     3.12   0.002     .0032149    .0141149
ln_Stadium_~y |          0  (omitted)
        _cons |  -.4688619   .1513337    -3.10   0.002    -.7659955   -.1717283
-------------+----------------------------------------------------------------
      sigma_u |  .11428993
      sigma_e |  .39904719
          rho |  .07581036   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(29, 685) = 1.12                     Prob > F = 0.3103
```

Figure 15: Running a Fixed Data Regression for "logwin" dependent variable.

Estimated Equation

$$\widehat{ln(Win\%)} = 0.1935 + 0.6106(PWP) + 0.0086(PYR) - 0.3998(SIT) - 0.0024(MS) + 0.0001(SR) \quad (4)$$

Playoffs Result Model

For the second regression, when creating the variable for Playoff result, the dependent variable becomes a dummy variable, which is either 1 if a team made the playoffs or 0 if they failed to do so. Using a dummy variable as the dependent variable required that the regression results be analyzed differently. Previously, a linear model had been used for the regression analysis; however, with a dummy dependent variable, a linear model does not work because it just clusters the data points around 1 and 0. To accurately analyze the data, a Binomial Logit model was used since the Logit model creates a curved motion between the points that better captures the true relationship of the dependent and independent variables.

```
. xtlogit PlayoffsResult StateIncomeTaxbase2010and MarketSizeinmillions PriorSeasonWin new_order_StadiumRank new_order_PayrollRank ln_Stadiu
> m_Capacity, fe
note: multiple positive outcomes within groups encountered.
note: ln_Stadium_Capacity omitted because of no within-group variance.

Iteration 0:  Log likelihood = -322.40741
Iteration 1:  Log likelihood = -319.62567
Iteration 2:  Log likelihood =  -319.5429
Iteration 3:  Log likelihood = -319.54288

Conditional fixed-effects logistic regression       Number of obs    =    720
Group variable: team_id                             Number of groups =     30

                                                    Obs per group:
                                                                  min =     24
                                                                  avg =   24.0
                                                                  max =     24

                                                    LR chi2(5)       =  25.37
Log likelihood = -319.54288                         Prob > chi2      = 0.0001
```

PlayoffsResult	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
StateIncomeTaxbase2010and	-2.897414	6.656405	-0.44	0.663	-15.94373	10.1489
MarketSizeinmillions	-.0654887	.0622774	-1.05	0.293	-.1875501	.0565727
PriorSeasonWin	2.06226	1.069526	1.93	0.054	-.033972	4.158493
new_order_StadiumRank	.0056487	.0361006	0.16	0.876	-.0651071	.0764045
new_order_PayrollRank	.0610752	.0170101	3.59	0.000	.027736	.0944144
ln_Stadium_Capacity	0	(omitted)				

Figure 16: Running a Binomial Logit Regression, accounting for a Fixed Data Regression for PlayoffsResults.

Estimated Equation

$$xL: \widehat{Pr(D_i} = 1) = \ -\ 2.897(SIT)\ -\ 0.065(MS)\ +\ 2.062(PWP)\ +\ 0.006(SR)\ +\ 0.061(PYR)$$

(5)

Model Interpretation

In this model, payroll rank is by far the most significant variable. Interpreting its coefficient, improving payroll ranking by one spot would increase a team's percent chance of making the playoffs by about 1.5% (0.061/4 = 0.01525). Moreover, it is considered statistically significant at a 95% confidence interval, with a p-value of 0.000 < 0.005. Prior season win now on the precipice, with a p-value of 0.054 > 0.005, and can be considered insignificant in this dataset. This variable may prove to be more significant than this dataset lets on, however, and should not be discounted as such. Conversely, variables related to market size, stadium rank, and state tax failed to reach statistical significance, indicating that these factors do not exert a reliable influence on playoff chances when controlling for team-specific effects.

Conclusion

This study analyzed whether externalities exert a measurable influence on competitive performance in Major League Baseball through two lenses: predicting regular-season record (xtreg logwin) and qualifying for the postseason (xtlogit PlayoffsResult). In prioritizing a fixed-effects specification for both models, a tangible portion of team-specific noise was effectively filtered out of the model, helping isolate how changes in the dependent variables influence success in the modern era.

The most consistent finding across each model is that success in the MLB is primarily a function of payroll rank and prior season wins. In the fixed-effects winning percentage model, payroll rank ($p = 0.002$) correlates an increase in team spending relative to its peers with higher win totals. In the model for playoff chances, this relationship is even more pronounced ($p = 0.000$), where moving up a single spot in payroll increases playoff likelihood by 1.5%. Regarding prior season wins, the correlation is less significant in the playoff model, yet still near the threshold for significance, indicating that the momentum of a successful roster/management is a strong predictor of making the playoffs. Assuming winning percentage and playoff result are at least somewhat correlated (having a majority of matching variable coefficients within this study between models would concur), this would indicate that making the playoffs in one year increases the chances of making the playoffs in the next, although additional research would be needed to support or deny this hypothesis.

Crucially, this research does not support any other significant explanatory variables within the playoff model. Most literature supported the notion that, with better stadiums, more favorable state taxes, and larger markets, teams should win more often. In the initial logwin model, coefficients aligned with the hypothesized, yet none of the variables were significant. Still, the theory stated they might hold relevance. In the more pertinent playoff model, these independent variables still hold no statistical significance, with an observed negative coefficient for market size (this is theorized as a plausible outcome within the variable description).

In the introduction of this paper, the effects of the Moneyball revolution are highlighted, which looked at finding competitive edges through efficiency over total spending. Similar to the events in the Moneyball film, the study's results support how teams have used econometrics to

maximize their chances of success, as well as interpreting general predictors of success. The results showed that season winning is predominantly influenced by prior season winning and payroll, while payroll is the stronger predictor for reaching the playoffs. Normally, this would lead to the conclusion that, in the modern MLB, dramatic spending remains a statistically sound strategy. While technically true, the low R-squared found within this study, as well as many sports studies, suggests that much of the variation in the determinant of success comes from variables impossible to include in the study's models; this is where sabermetrics excels.

Overall, while baseball is a game characterized by internal metrics, the external framework is still largely predicted through money spent. Physical infrastructure and geographic location are less important for success than a franchise's willingness to invest in its team. This is especially applicable to making the playoffs, where the explanatory power of team payroll is proportionally more significant. Even so, the inherent noise of the game and the compressed nature of MLB win totals indicate that efficiency and internal externalities continue to play larger roles in maintaining competitive balance.

List of References

Link to study's excel sheet: (WPI view only access) Econometrics Final Paper Data Sheet .xlsx

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Appendix

I – Regression Model I:

Pythagorean as the Dependent Variable

```
. xtreg Pythagoreanwin StateIncomeTaxbase2010and MarketSizeinmillions PriorSeasonWin new_order_StadiumRank new_order_PayrollRank ln_Stadium_
> Capacity, fe
note: ln_Stadium_Capacity omitted because of collinearity.

Fixed-effects (within) regression               Number of obs      =       720
Group variable: team_id                         Number of groups   =        30

R-squared:                                      Obs per group:
     Within  = 0.1111                                         min =        24
     Between = 0.6047                                         avg =      24.0
     Overall = 0.2120                                         max =        24

                                                F(5, 685)          =     17.13
corr(u_i, Xb) = 0.2005                          Prob > F           =    0.0000

------------------------------------------------------------------------------
Pythagoreanwin | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
---------------+--------------------------------------------------------------
StateIncomeTax~d |  -.1512143   .1823235    -0.83   0.407    -.5091943    .2067656
MarketSizeinmi~s |   .0010032   .0016636     0.60   0.547    -.0022632    .0042697
PriorSeasonWin |   .1735733   .0284603     6.10   0.000     .1176934    .2294533
new_order_Stad~k |  -.0000771   .0009056    -0.09   0.932    -.0018552    .0017011
new_order_Payr~k |   .0021783   .0004668     4.67   0.000     .0012618    .0030949
ln_Stadium_Cap~y |          0  (omitted)
           _cons |   .3876777   .0254513    15.23   0.000     .3377058    .4376496
---------------+--------------------------------------------------------------
         sigma_u |  .02423382
         sigma_e |  .0671117
             rho |  .1153503   (fraction of variance due to u_i)
------------------------------------------------------------------------------
F test that all u_i=0: F(29, 685) = 2.30                     Prob > F = 0.0001
```

Figure 18: Running a Fixed Data Regression with Pythagorean Win % as the dependent variable.

II – Regression Model II:

Regressing–regular season win percentage as the dependent variable against the independent variables payroll rank, state income tax, market size, prior season win percentage, and Pythagorean win percentage–the test found an adjusted R-squared of 0.4169, with the caveat that payroll rank and state income tax now become insignificant. Essentially, the Pythagorean win percentage is so significant in this model that other explanatory variables are essentially drowned out. This, along with the Pythagorean win percentage being a more "sabermetric" approximation based on actual performance, would cause this variable to be detrimental in the study's model.

Estimated Equation

$$RegularSeasonWin\% = B_0 + B_1 * PYR + B_2 * SIT + B_3 * PSW + B_4 * SR + B_5 * PythagoreanWin\%$$

(6)

Where:

- $RegularSeasonWin\%$, between 0 and 1

- PYR, 30 being the highest-ranked team and 1 being the lowest
- SIT, base 2010, between 0 and 1
- PSW, between 0 and 1
- SR, with 30 being the highest-ranked and 1 the lowest

$$PriorSeasonWin\% = \frac{Regular\ Season\ Wins}{162}$$

- $PythagoreanWin\%$, between 0 and 1

$$PythagoreanWin\% = \frac{(Runs\ Scored)^2}{(Runs\ Scored)^2 + (Runs\ Allowed)^2}$$

```
. regress regseasonwin payrollrank stateincometaxbase2010andassumed marketsizein
> millions priorseasonwin pythagoreanwin
```

Source	SS	df	MS	Number of obs	=	720
				F(5, 714)	=	103.83
Model	2.80354424	5	.560708849	Prob > F	=	0.0000
Residual	3.85586716	714	.005400374	R-squared	=	0.4210
				Adj R-squared	=	0.4169
Total	6.6594114	719	.009262046	Root MSE	=	.07349

regseasonwin	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
payrollrank	-.0003164	.0003912	-0.81	0.419	-.0010844	.0004516
stateincome~d	-.0004194	.0793771	-0.01	0.996	-.1562598	.1554211
marketsizei~s	.0004526	.0006234	0.73	0.468	-.0007713	.0016764
priorseason~n	.0337691	.0313527	1.08	0.282	-.0277854	.0953236
pythagorean~n	.7578437	.0397087	19.09	0.000	.6798839	.8358035
_cons	.0924083	.0252982	3.65	0.000	.0427406	.1420761

Figure 19: Running a Fixed Data Regression where the dependent variable is the percentage of wins and the Pythagorean Win Percentage is a dependent variable

This regression is "complete" in the sense that it accounts for both external factors, such as market size and payroll, while taking into account internal factors like runs score and runs allowed. This is proven through the high adjusted R-squared, which is more than double that of comparable studies that did not account for internal factors. To note, variables such as the strength of schedule were not accounted for due to data scarcity and time constraints around data scraping.

III – Additional Data Visualization:

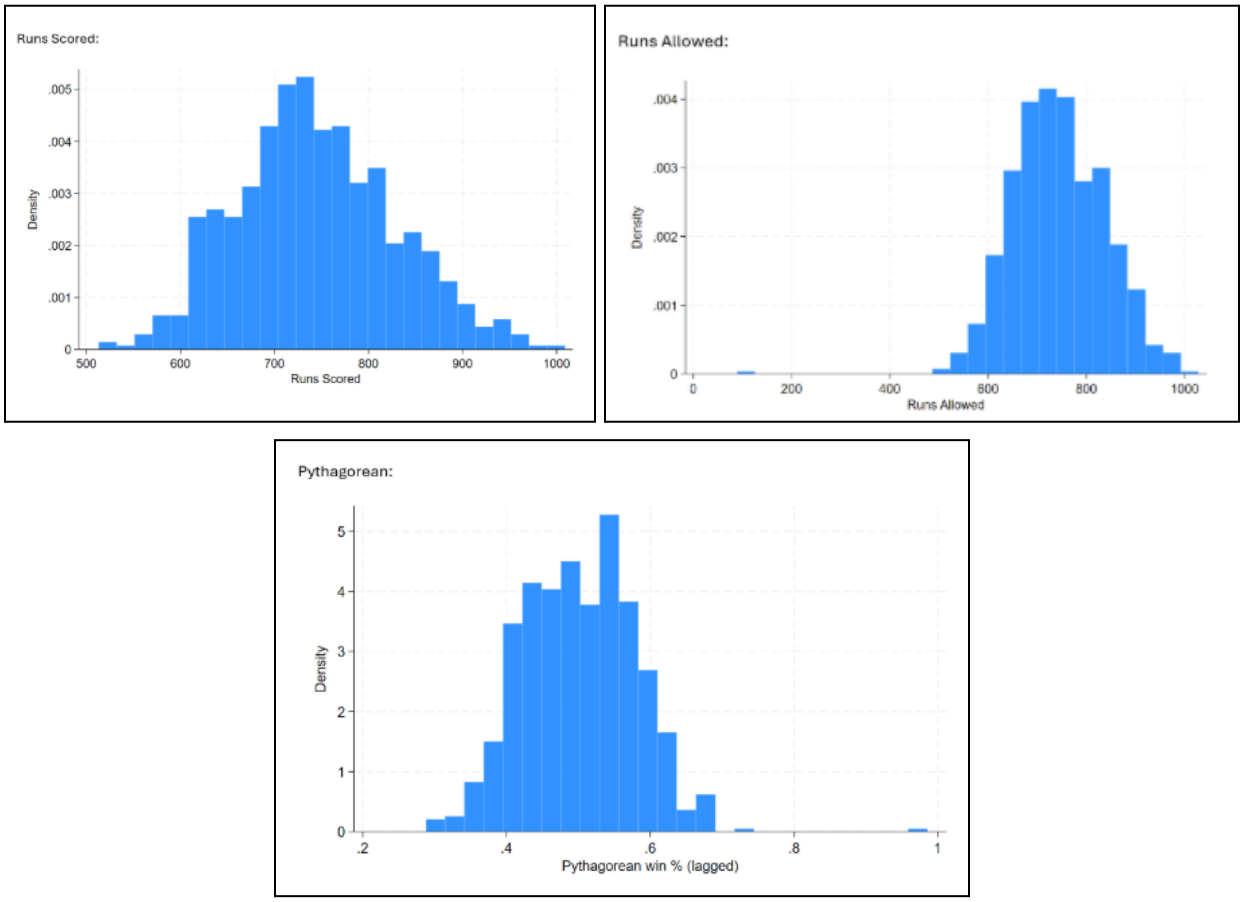


Figure 20: Histograms for Runs Scored, Runs Allowed, and Pythagorean Win %. These values were not included in the actual study but were interesting to analyze.

IV – Potential biases:

- Year fixed effects — the 1998 to 2022 window this study covered saw the game of baseball change dramatically. Times such as the steroid era, Moneyball diffusion, and analytics revolution changed the way of the game in ways that are difficult to account for in regressions.
- Division fixed effects — the two leagues, the American League and National League, often vary in difficulty and hence this often restricts playoff-worthy teams from making the playoffs. For example, if 8 of the top 10 teams are in the American League only 6 of them can make it to the playoffs as each league has 6 playoff slots. As well, within leagues themselves there are structural differences that make it harder.

- Number of playoff spots — before 2012 the postseason only included 8 teams, with 1 wildcard per league. Yet this changed in 2012 with the addition of an extra wildcard slot per league, expanding the postseason to 10 teams.
- Payroll inflation — the average MLB salary has increased faster than inflation due to the expansion of the MLB market. This can cause the payroll of more recent seasons to be overweighted and become biased.
- Stadium rank — variables such as this are categorically subjective and are prone to bias.

IIV – t and z Test:

t-Test for logwin at 5%:

t-Test for State Income Tax:

$Ho = 0$

$Ha < 0$

$t_c = -1.96$

$t = -1.39$

$insignificant: -1.39 > -1.96$

t-Test for Prior Season Win %:

$Ho = 0$

$Ha > 0$

$t_c = 1.96$

$t = 3.63$

$significant: 3.63 > 1.96$

t-Test for ln(Stadium Capacity):

$Ho = 0$

$Ha > 0$

$t_c = 1.96$

$t = omitted$

$insignificant$: $variable\ was\ omitted$

t-Test for Stadium Rank:

$Ho\ =\ 0$

$Ha\ >\ 0$

$t_c\ =\ 1.96$

$t\ =\ 0.06$

$insignificant$: $0.06\ <\ 1.96$

t-Test for Payroll Rank:

$Ho\ =\ 0$

$Ha\ >\ 0$

$t_c\ =\ 1.96$

$t\ =\ 3.12$

$significant$: $3.12\ >\ 1.96$

t-Test for Market Size (two tail test):

$Ho\ =\ 0$

$Ha\ >\ 0$

$Ha\ <\ 0$

$t_c\ =\ 1.96$

$t\ =\ 0.96$

$insignificant$: $|0.96|\ <\ |1.96|$

z-Test for PlayoffsResult at 5%:

z-Test for State Income Tax:

$Ho\ =\ 0$

$Ha\ <\ 0$

$z_c\ =\ -\ 1.96$

$z = \ -0.44$

$insignificant$: $-0.44 > -1.96$

z-Test for Prior Season Win %:

$Ho = 0$

$Ha > 0$

$z_c = 1.96$

$z = 1.93$

$insignificant$: $1.93 < 1.96$

z-Test for ln(Stadium Capacity):

$Ho = 0$

$Ha > 0$

$z_c = 1.96$

$z = omitted$

$insignificant$: $variable\ omitted$

z-Test for Stadium Rank:

$Ho = 0$

$Ha > 0$

$z_c = 1.96$

$z = 0.16$

$insignificant$: $0.16 < 1.96$

z-Test for Payroll Rank:

$Ho = 0$

$Ha > 0$

$z_c = 1.96$

$z = 3.59$

$significant$: $3.59 > 1.96$

z-Test for Market Size (two tail test):

$Ho = 0$

$Ha > 0$

$Ha < 0$

$z_c = |1.96|$

$z = -1.05$

$insignificant$: $|-1.05| < |1.96|$

IIIV – Image of Do-File of Commands:

```
histogram PriorSeasonWin
twoway (scatter logwin PriorSeasonWin) (lfit logwin PriorSeasonWin)
browse
preserve
regress logwin StateIncomeTaxbase2010and MarketSizeinmillions PriorSeasonWin new
> _order_StadiumRank new_order_PayrollRank ln_Stadium_Capacity
est sto m1
estout
xtreg logwin StateIncomeTaxbase2010and MarketSizeinmillions PriorSeasonWin new_o
> rder_StadiumRank new_order_PayrollRank ln_Stadium_Capacity, fe
est sto m2
estout m1 m2
estat hettest
vif
corr logwin StateIncomeTaxbase2010and MarketSizeinmillions PriorSeasonWin new_or
> der_StadiumRank new_order_PayrollRank ln_Stadium_Capacity
xtset
xtlogit PlayoffsResult StateIncomeTaxbase2010and MarketSizeinmillions PriorSeaso
> nWin new_order_StadiumRank new_order_PayrollRank ln_Stadium_Capacity, fe
```